SENG272 Project Report:

Requirements & Design

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Course: SENG272 Software Project II

Date: March 29, 2026

1. Project Overview

The objective of this project is to develop a high-fidelity Java Swing desktop application that simulates the **ISO/IEC 15939** software measurement process. The application follows a 5-step structured "Wizard" workflow, enabling users to define quality dimensions, collect raw data, and perform automated gap analysis through advanced data visualization.

2. Requirements Specification

2.1. Functional Requirements

- **User Identification:** Capturing and validating academic profile details (Username, School, Session).
- **Scope Definition:** Selection of Quality Type (Product/Process) and Operational Mode (Health/Education) using mutually exclusive controls.
- **Measurement Planning:** Automated retrieval of scenario-specific metrics and their respective coefficients.
- **Automated Scoring & Analysis:** Calculation of 1.0–5.0 scores based on ISO-compliant formulas and identifying performance gaps.

2.2. Non-Functional Requirements

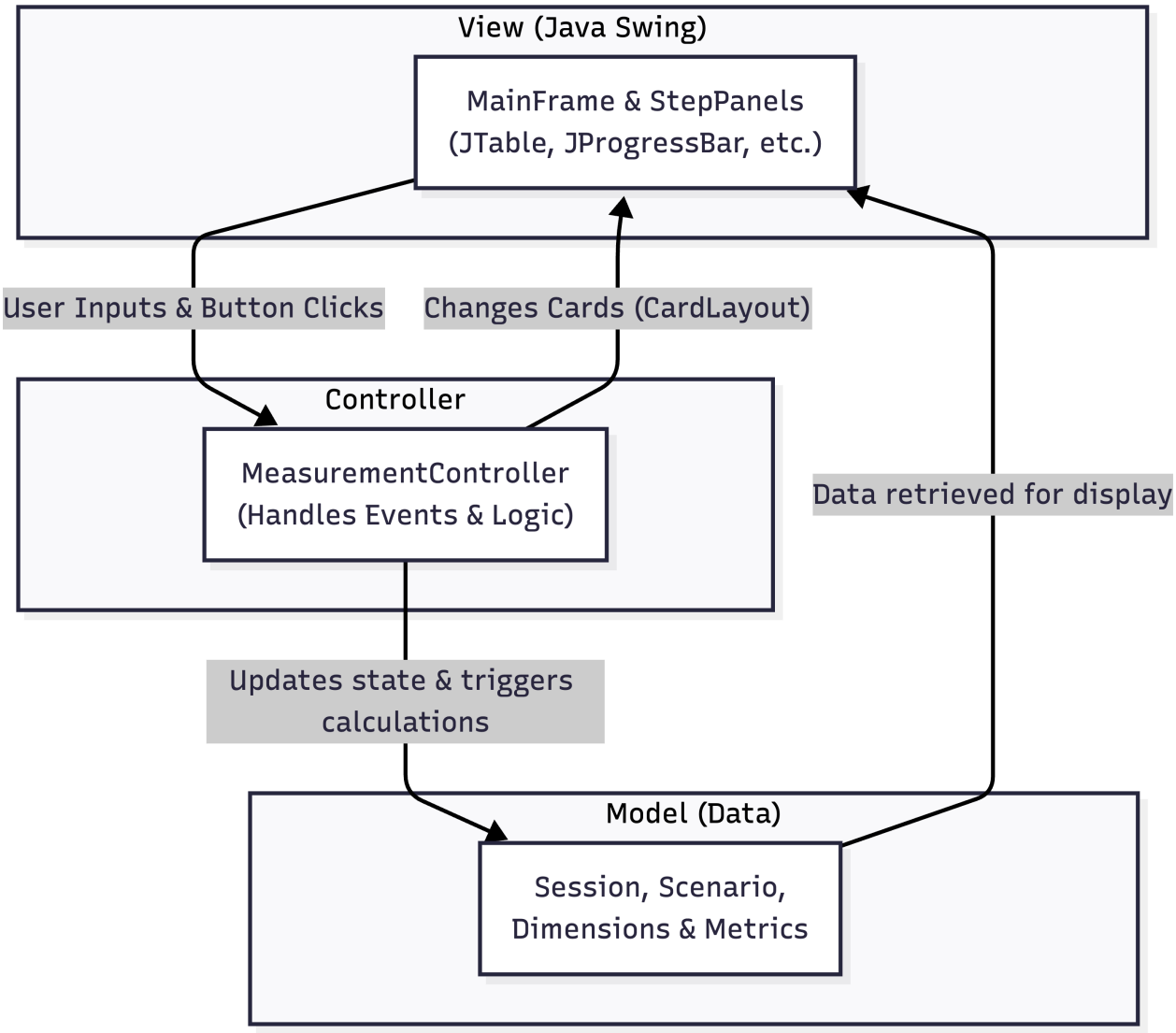
- **Architecture:** Strict adherence to the **MVC (Model-View-Controller)** pattern.
- **Usability:** A consistent "Wizard" navigation system with active-step highlighting and progress tracking.
- **Technical Stack:** Java SE 17+ utilizing standard Swing and Collections libraries.

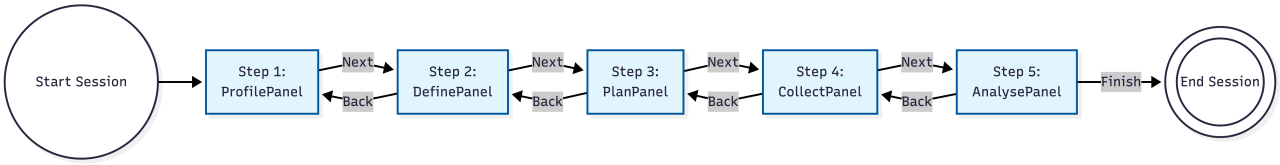
3. System Design & Architecture

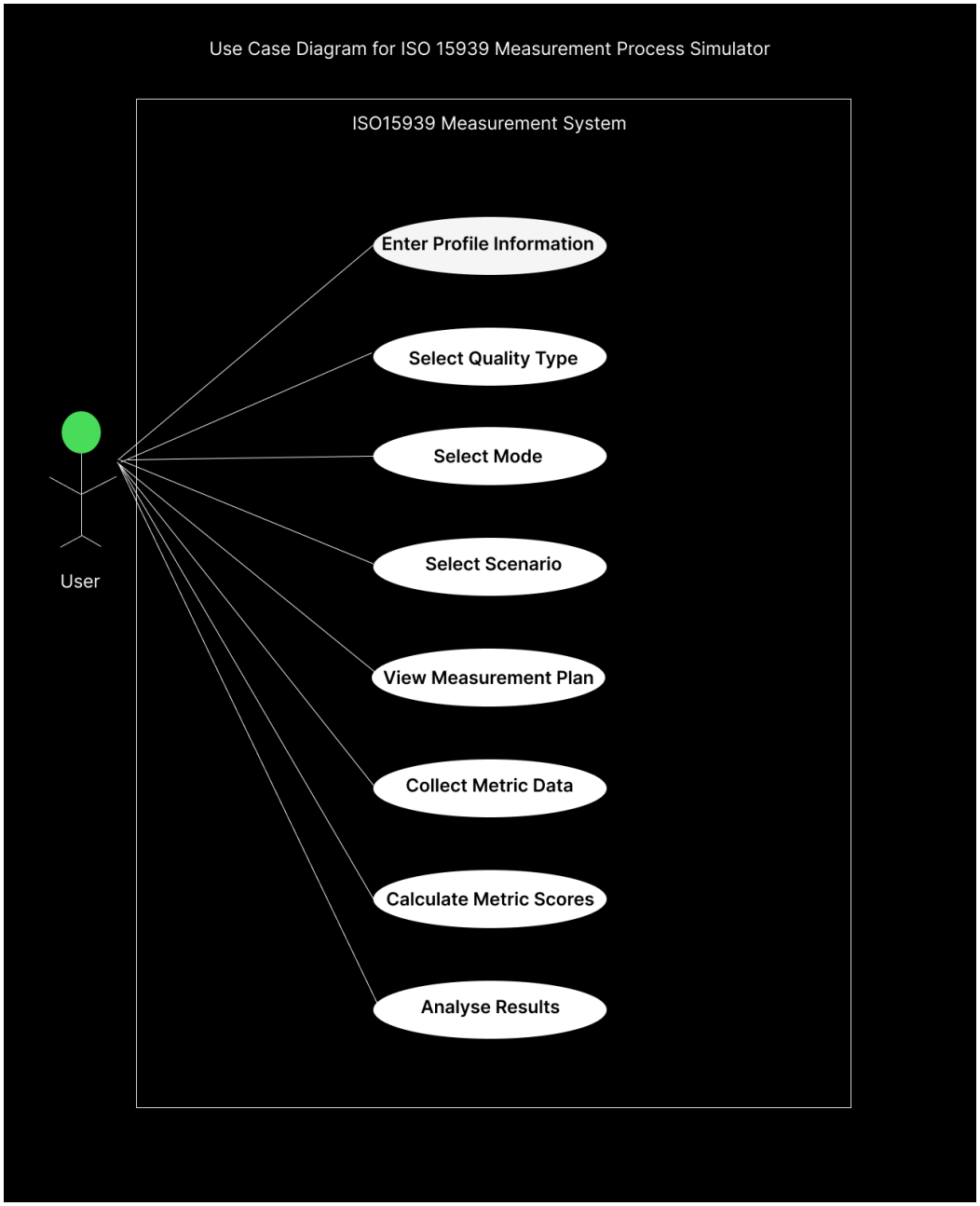
3.1. Architectural Design

The system architecture ensures a clean separation between data management and user interface.

- **Model:** Encapsulates `Session`, `Scenario`, `Dimension`, and `Metric` data structures.
- **View:** Manages the visual components using a `CardLayout` for seamless step transitions.
- **Controller:** Facilitates event handling and state updates between the Model and View.

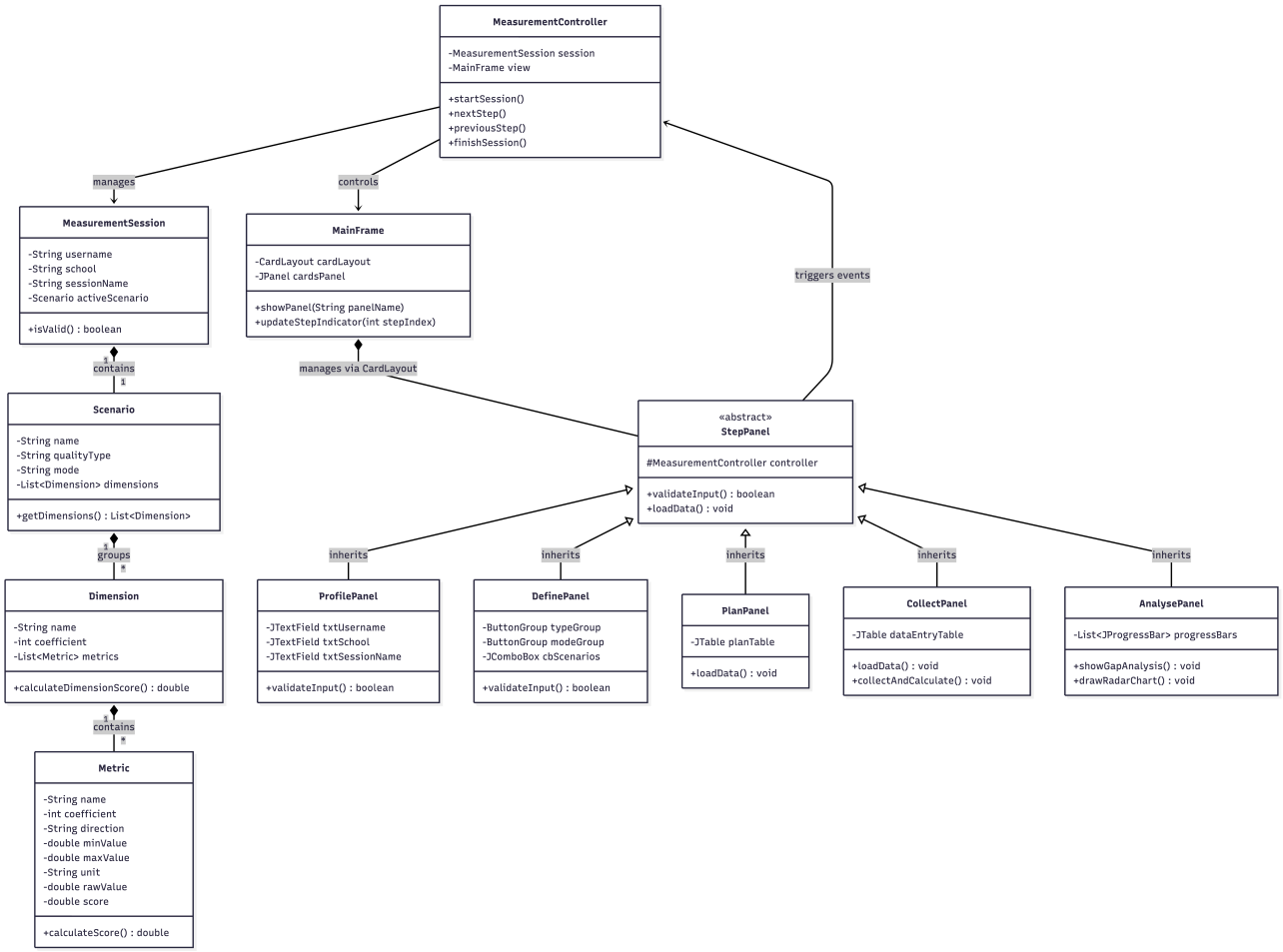



Start Session
Step 1: ProfilePanel
Next
Back
Step 2: DefinePanel
Next
Back
Step 3: PlanPanel
Next
Back
Step 4: CollectPanel
Next
Back
Step 5: AnalysePanel
Finish
End Session


Use Case Diagram for ISO 15939 Measurement Process Simulator
ISO15939 Measurement System
Enter Profile Information
Select Quality Type
Select Mode
Select Scenario
View Measurement Plan
Collect Metric Data
Calculate Metric Scores
Analyse Results
User

3.2. Class Hierarchy & OOP Principles

The implementation utilizes an abstract `StepPanel` class, which serves as a blueprint for all five stages of the wizard. Complex data relationships are managed through `ArrayList` and `HashMap` collections to support dynamic metric grouping.

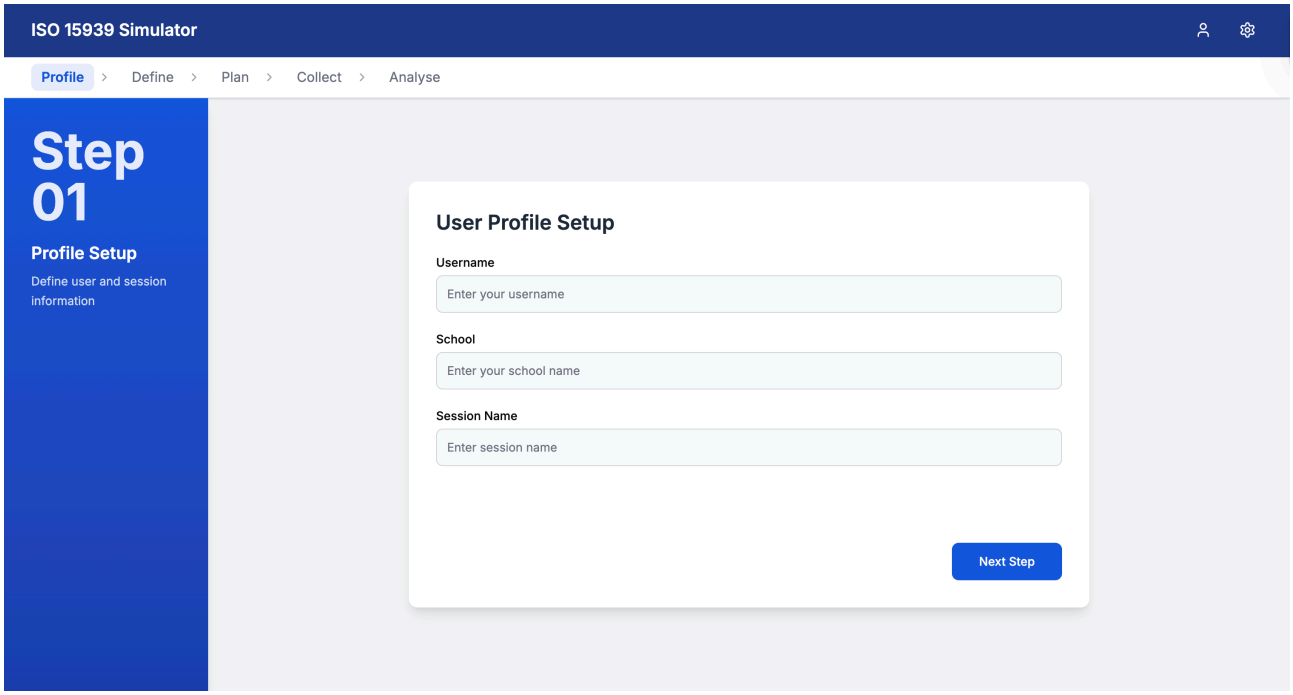


4. Graphical User Interface (GUI) Mockups

The following mockups represent the high-fidelity Figma design for the application. The design maintains **pixel-perfect consistency** across all stages, ensuring that global navigation elements remain static while the central content card updates dynamically.

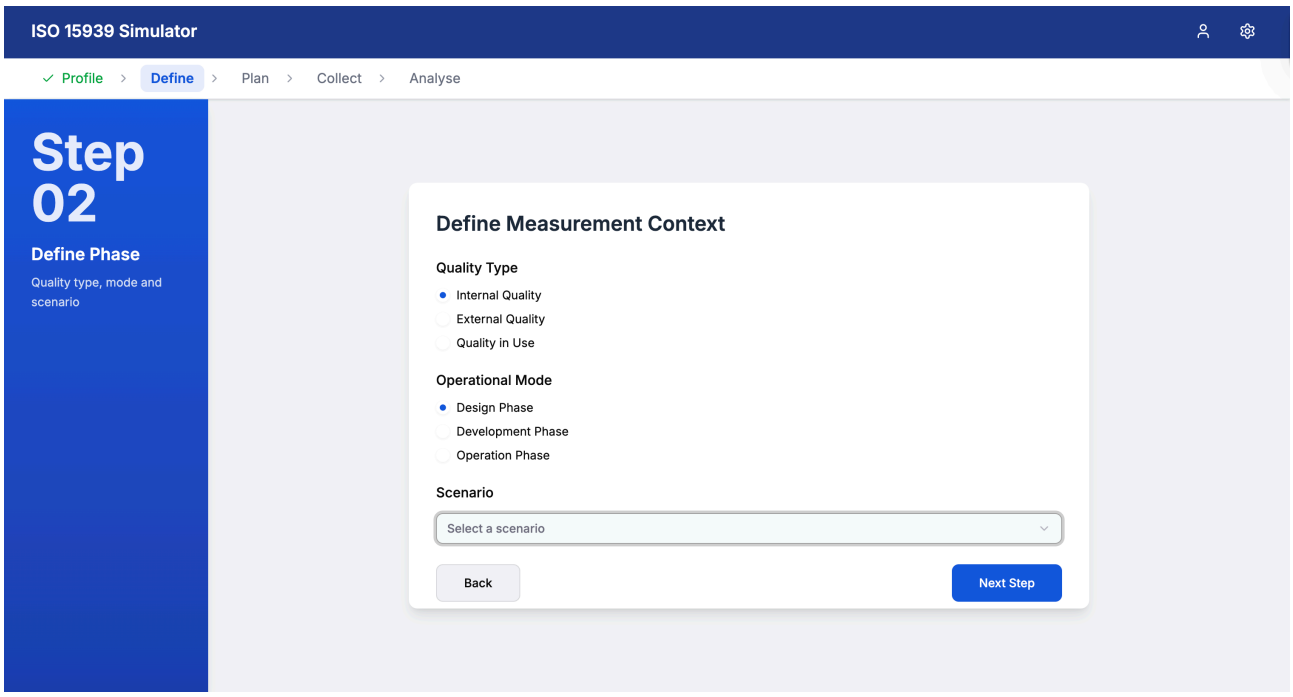
Step 1: User Profile Configuration

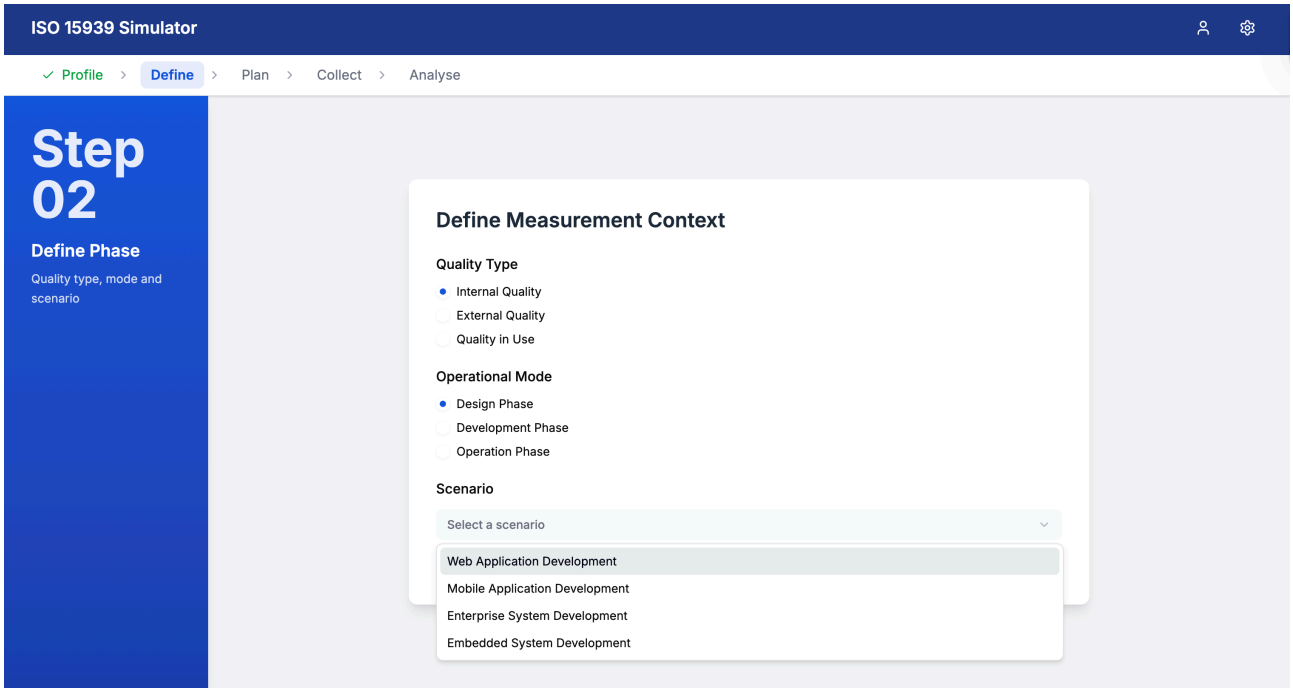
Initial setup stage for capturing simulation credentials.



Step 2: Define Quality Dimensions (Scope)

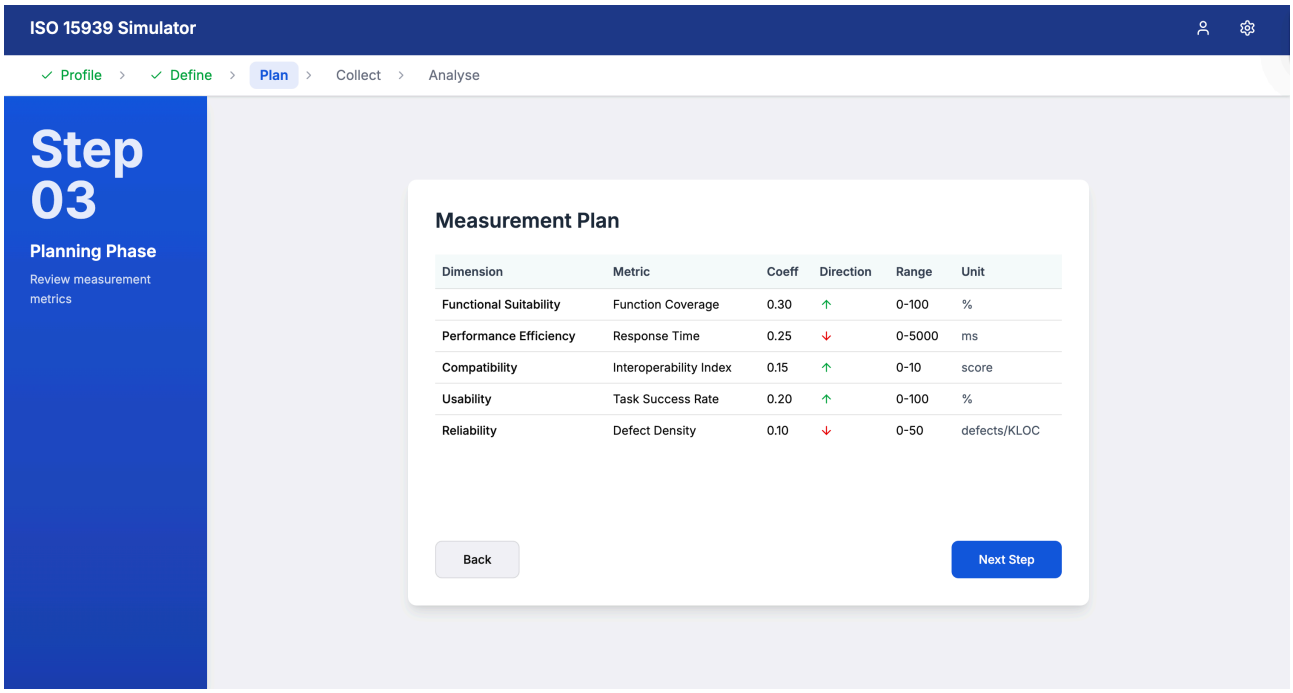
Selection of measurement parameters using JRadioButtons and JComboBox components.





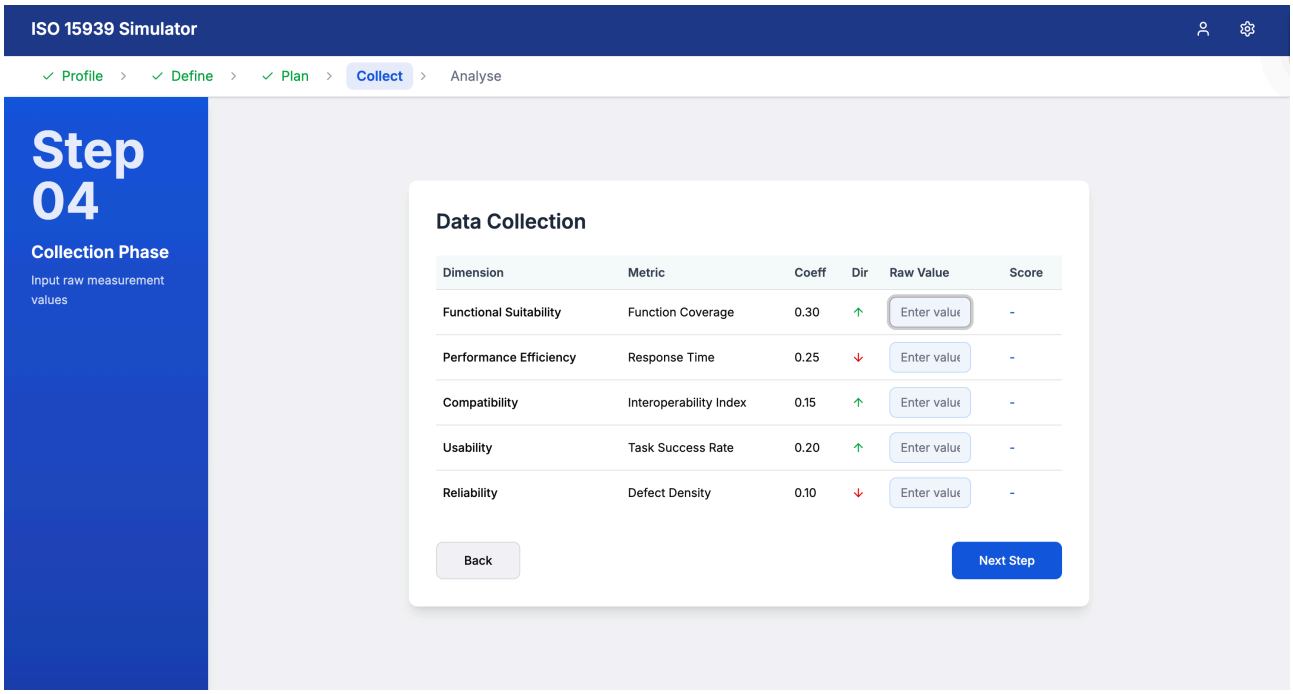
Step 3: Measurement Plan (Read-Only Data)

Presentation of scenario-specific metrics in a structured, non-editable JTable format.



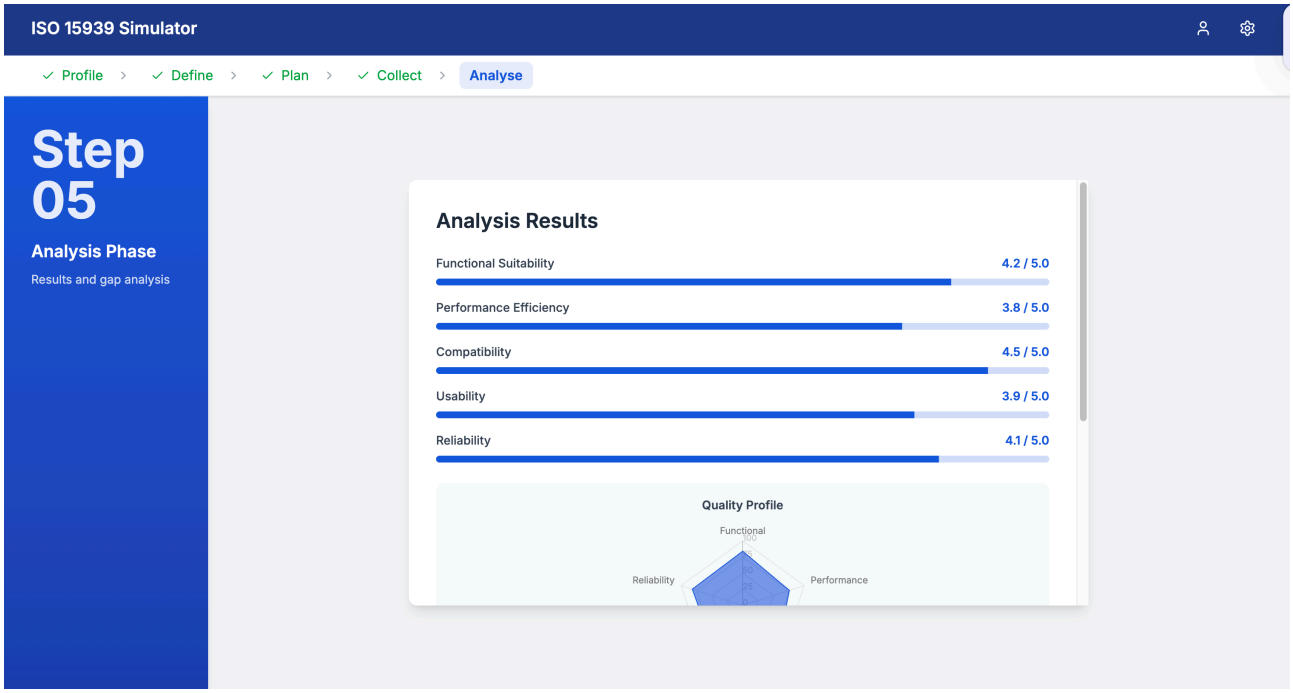
Step 4: Data Collection (Active Input)

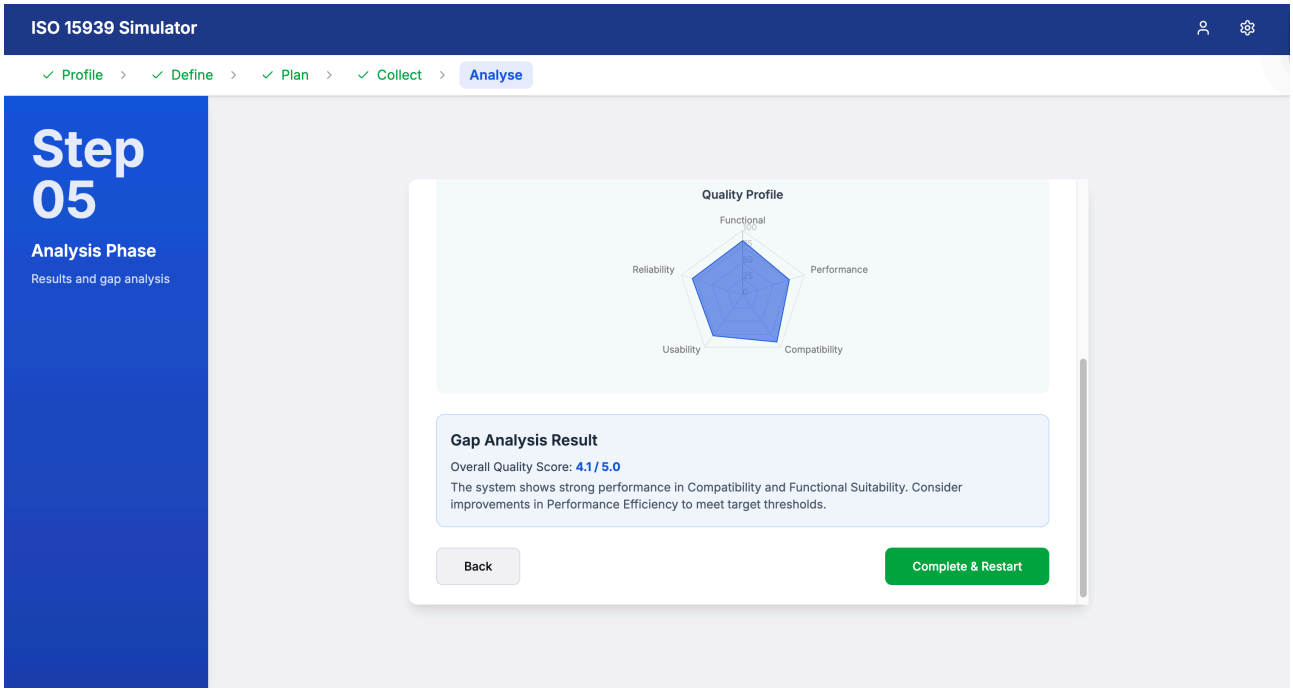
Interactive interface for raw data entry with automated score calculation.



Step 5: Continuous Improvement Analysis

Final dashboard featuring **JProgressBars**, a **Radar Chart**, and a **Gap Analysis Summary** for performance evaluation.





5. Technical Disclaimer on Repository History

Important Note: Due to a technical Git synchronization error (force-push) during repository reorganization on March 29, the commit history for HW1 and HW2 was reset. This report confirms that all project deliverables were originally completed and submitted in accordance with the course schedule.

6. Conclusion

This design documentation provides a robust foundation for the implementation of the ISO 15939 Simulator. By combining an MVC architecture with a high-fidelity "Wizard" interface, the project is positioned to meet all technical and functional objectives of the course.